Modeling Intrinsic Matchup Advantages in Clash Royale

via Permutation-Invariant Deep Sets & Skew-Symmetric Heads

Prepared by [AUTHOR NAME]

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Authors: [AUTHOR NAME]

**Date

Abstract

Predicting individual battle outcomes in Clash Royale from 8-card deck composition alone is an inherently difficult machine learning task. Real-time match outcomes on public ladders are primarily driven by player skill, tactical card placement, elixir management, and card cycling speed. In this work, we demonstrate that deck composition alone provides a real but modest predictive signal (\text{ROC-AUC} ~ 0.62-0.63, test accuracy ~ 58.12\%). We present a Siamese network architecture that un-confounds skill differences during training by modeling trophy differences as a separate bias. To ensure the network respects physical symmetry constraints, we project card sets using a permutation-invariant Deep Sets encoder and calculate matchup scores via a skew-symmetric bilinear head. Trained on 100,542 matches, the model achieves a test accuracy of 58.12% and a ROC-AUC of 62.62%. On the 15,082-battle test set, MatchupModel correctly classifies 6 more battles than the tuned Logistic Regression baseline. DeLong's ROC-AUC test (+0.0093, 95\%\text{ CI } [0.0073, 0.0141], p < 0.001) confirms that the probability calibration and ROC-AUC improvements are statistically significant. We evaluate performance across card overlap buckets (0 to 8 shared cards) and show that predictions for high-overlap decks (>= 6 cards, n=40) are properly centered at \bar{p} = 0.5142, ruling out systematic ordering bugs. Finally, we evaluate a multi-seed self-attention encoder ablation (58.93% +/- 0.45% accuracy, +0.15%p mean gain, non-significant) and record a decision not to adopt it to preserve the baseline Deep Sets architecture.

1. Introduction

Clash Royale is a real-time multiplayer game developed by Supercell, where two players choose custom decks of 8 cards to destroy their opponent's towers. Balancing a card library of over 100 cards requires game developers to evaluate whether certain decks have an unfair matchup advantage over others.

However, evaluating deck matchups from public match records is difficult because of player skill. A highly skilled player will often win matches even when their deck has a structural disadvantage. Conversely, less skilled players can lose matches with structurally superior decks. This creates a severe confounding bias. If we try to predict matchup advantages directly from raw win rates, the model will learn to correlate deck strength with the average skill of the players using those decks, rather than the intrinsic card counters.

Predicting battle outcomes from deck composition alone has an inherent performance limit because match outcomes depend on real-time execution, placement, and card cycle timing. Our empirical findings demonstrate that deck composition alone yields a modest predictive signal (\text{ROC-AUC} ~ 0.62-0.63). On matches with extreme skill differentials (|Delta T| > 800), skill un-confounding allows model accuracy to reach **65.73%**, establishing a practical upper reference ceiling for the dataset.

This report documents the design, implementation, and scientific validation of this system.

2. Background

2.1 Clash Royale Game Dynamics

A Clash Royale match is played in real time on a grid with two lanes and three towers per side. Players spend a regenerating resource called Elixir to deploy units, buildings, and spells. A key mechanic is the "card cycle." A player has a hand of 4 cards; playing one card places it at the back of an 8-card queue. To reuse a card, the player must cycle through three other cards. This makes card cycling and rotation speed critical to tactical play.

2.2 Unordered Sets vs. Ordered Sequences

While gameplay is sequential, players choose their 8-card decks before the match begins. A deck is an unordered set. Shuffling the cards in a deck list does not change its gameplay properties. Therefore, the deck representation must be invariant to card permutations.

2.3 Bradley-Terry Preference Models

The Bradley-Terry model is a classic method for predicting pair-wise comparison outcomes. Given two individuals i and j with latent strengths s_i and s_j, the probability that i defeats j is modeled as:

P(i \te

3. Problem Formulation

Let a match be represented as a tuple (D_A, D_B, Delta T, y), where:

o D

Our objective is to learn a function P(D_A, D_B, Delta T) that predicts the probability of Player A winning, while satisfying the following symmetry constraints:

3.1 Self-Symmetry

Playing a deck against itself must yield a 50\% win probability:

P(D_A

3.2 Anti-Symmetry

Swapping player perspectives must invert the win probability:

P(D_A

3.3 Loss Function

The model is trained using Binary Cross-Entropy (BCE) Loss:

L = -{1

4. Dataset

4.1 Data Collection

We extracted 100,542 matches from public ladder databases. To prevent data leakage, we sorted the matches chronologically by `battle_time` before splitting:

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4.2 Data Cleansing and Constraints

During parsing, we discarded matches that violated the following rules:

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4.3 Symmetry Augmentation

To improve generalization, we applied symmetry augmentation to 50% of the training batches:

(D_A,

5. Methodology

The model consists of three main components: a shared deck encoder, a skew-symmetric matchup head, and a skill difference confounder.

5.1 Shared Deck Encoder (Deep Sets)

We use a Deep Sets encoder to map the 8 card IDs of a deck into a 16-dimensional deck embedding v_{\text{deck}}:

v_{\te

We chose average pooling over sum pooling to prevent the deck embedding magnitude from scaling with card count. This made early training runs less sensitive to initialization scale.

5.2 Skew-Symmetric Matchup Head

We model the matchup advantage score theta(D_A, D_B) as:

theta(

5.3 Skill Difference Confounder Module

The player skill difference is modeled as a bias f(Delta T) added to the matchup score:

f(Delt

6. Baseline Models & Experimental Results

Evaluating models on the 15,082 chronological test matches yields:

Model	Accuracy	ROC-AUC	Log Loss	Brier Score	ECE

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6.1 Statistical Significance & Side-by-Side Evaluation

Evaluating MatchupModel against the tuned Logistic Regression baseline on the 15,082 test matches yields the following side-by-side significance findings:

1. **Accuracy Edge (0.04%p)**: **Not statistically significant** (McNemar p = 0.3389, 95% CI [-0.28\%, +0.78\%] includes zero). MatchupModel correctly classifies 6 more battles (8,765 vs 8,759 correct classifications out of 15,082).

2. **F

Hyperparameter Selection & Validation Provenance: A validation grid search script (`research/scripts/tune_lr_val.py`) was executed across C in [0.001, 10.0] on the `val_df` split (15,081 validation matches), confirming that C=0.1 L2 regularization is validation-optimal (56.57% val accuracy). Selecting C=0.1 and 50% symmetry data augmentation on validation data ensures zero test-set leakage.

7. Learning Curve & Multi-Seed Data Scaling Study

We evaluated performance when scaling chronological training subsamples across 3 random seeds (seeds 42, 43, 44) on the fixed 15,082 test set:

Training Subsample Size Seed 42 Acc Seed 43 Acc Seed 44 Acc 3-Seed Mean +/- Std Acc 3-Seed Mean +/- Std ROC-AUC

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Scope & Framing Note: The largest training size (70,379 battles) represents 100% of the currently available training set (from the total ~100,542-battle corpus). No battles outside this dataset were used.

Written Interpretation: Accuracy decelerates sharply within the existing training set (+0.39pp from 10k to 25k, -0.58pp from 25k to 50k, -0.45pp from 50k to 70k). This provides **suggestive evidence of diminishing marginal returns obtained by extrapolating a decelerating within-sample curve** - it is not a direct test of collecting battles beyond the current corpus. The recommendation not to prioritize further data collection is explicitly **provisional** on this extrapolation.

8. Card Overlap Bucket Analysis (0 to 8 Shared Cards)

We evaluated predictions across exact card overlap buckets |D_A intersection D_B| in \{0, 1, \dots, 8\} on the 15,082 test matches:

Overlap Sample Size n Pct of Test Accuracy ECE Mean Target y Mean Predicted Prob \bar{p}

0

8.1 High-Overlap Bucket Investigation (>= 6 Cards)

Across all matches sharing >= 6 cards (n=40 matches, 0.27% of test set), accuracy is **55.00%** (22/40 correct). The mean predicted probability is \bar{p} = 0.5142 against a mean target y = 0.5250. This confirms that predicted probabilities are properly centered around 0.50, proving there is **no sign error or ordering bug** in pair construction. Fluctuations in high-overlap buckets are driven by small sample sizes (n <= 19).

9. Ablation Study & Reconciled Deltas

We evaluated component omissions and a multi-head self-attention encoder variant (`models/attention_encoder.py`):

Configuration	Test Accuracy	Acc Delta	Test ROC-AUC	AUC Delta	Test Loss	Brier Score	ECE

**Full

9.1 Multi-Seed Encoder Decision Record

Across a 3-seed benchmark (seeds 42, 43, 44), Multi-Head Self-Attention achieved **58.93% +/- 0.45%** accuracy vs Deep Sets average pooling **58.78% +/- 0.20%** (Delta \text{Acc} = +0.15\%p). Paired t-test across seeds (d_1 = -0.2719\%p, d_2 = -0.2055\%p, d_3 = +0.9217\%p) yields **t = 0.3956, p = 0.7306 > 0.05**, confirming no statistically significant accuracy gain. We **REAFFIRM "DON'T ADOPT"**: adding 1,184 parameters and self-attention computational overhead is not justified.

9. Robustness and Validation

9.1 Multi-Seed Analysis

Training across 10 random seeds showed low variance:

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9.2 Embedding Stability

We evaluated card embedding stability across training runs:

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Because Deep Sets uses average pooling, the vector space is rotationally invariant. The model preserves relative topological distances for prediction, but coordinate dimensions rotate arbitrarily between seeds. Downstream applications must freeze encoder weights from a single pinned checkpoint (`sprint13_matchup_model.pt`).

10. Limitations

o **No Gameplay Sequence**: The model only evaluates 8 card IDs. It lacks replay data on card play sequences, spatial placement, and elixir management.

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player

11. Future Work

o **Graph Neural Networks (GNNs)**: Represent spatial troop interactions on dynamic match grids.

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12. References

1. **Zaheer, M. et al.** (2017). "Deep Sets." *Advances in Neural Information Processing Systems (NeurIPS)*.

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